Chapter 17 exercises

Q1. Practice exercise page 721

Calculate the pH of a solution containing 0.085 M nitrous acid HNO_2 ; Ka = 4.5×10^{-4}) and 0.10 M potassium nitrite $NaNO_2$.

Answer:

HNO_2 is a weak acid

$NaNO_2$ is a salt of strong electrolyte

$HNO_2 \leftrightarrow H^+ + NO_2^-$

$NaNO_2 \rightarrow Na^+ + NO_2^-$

	HNO_2 $\leftrightarrow$	H^+ +	NO_2^-
initial	0.085 M	0 M	0 M
change	- X M	+ X M	+ X M
equilibrium	(0.085 – X) M	X M	(0.10 + X) M

$Ka = 4.5 \times 10^{-4} = [H^+] [NO_2^-] / [HNO_2] = (X) (0.1 + X) / (0.085 – X)$

$4.5 \times 10^{-4} = 0.10 X / 0.085$

$X = 3.825 \times 10^{-4}$ $[H^+]$ $pH = - \log [H^+] = - \log (3.825 \times 10^{-4})$

$pH = 3.42$

Q2. Practice exercise page722

Calculate the formate ion concentration and pH of a solution that is 0.050 M in formic acid (CHOOH) (Ka = 1.8×10^{-4}) and 0.10 M HNO_3 .

Answer:

$HCOOH_{(aq)} \leftrightarrow HCOO^-_{(aq)} + H^+_{(aq)}$

$HNO_{3\,(aq)} \rightarrow H^+ + NO_3^-$

	$HCOOH_{(aq)}$ $\leftrightarrow$	$HCOO^-_{(aq)}$ +	$H^+_{(aq)}$
initial	0.050 M	0 M	0.10 M
change	- X M	+ X M	+ X M

equilibrium (0.050 – X) X M (0.10 + X)

$Ka = 1.8 \times 10^{-4} = [HCOO^-] [H^+] / [HCOOH] = (0.10 + X) (X) / (0.050 – X)$

$X = 0.9 \times 10^{-4}$ M $[H^+]$

Concentration of $[H^+]$ after equilibrium $= 0.10 + X = 0.10 + 0.9 \times 10^{-4} = 0.10$ M

pH = - log (0.10) = 1.00

Q3. Practice exercise page 725

Calculate the pH of a buffer composed of 0.12 M benzoic acid and 0.20 M sodium benzoate.

($Ka = 6.3 \times 10^{-5}$)

Answer:

pH = pKa + log (base) / (acid)

pKa = - log ($6.3 \times 10^{-/5}$) = 4.2

pH = 4.20 + log (0.2 / 0.12) = 4.20 + 0.22 = 4.42

Q4. Practice exercise page 728

Determine:

a) The pH of the original buffer described in exercise 17.5 after the addition of 0.020 M HCl and

b) The pH of solution that would result from the addition of 0.020 M HCl to 1.00 L of pure water.

Answer:

a)	CH_3COOH ↔	CH_3COO^- +	H^+
initial	0.30 M	0.30 M	0 M
change	-	0 M	0.02 M
equilibrium	(0.30 + 0.02) M	(0.30 – 0.02) M	0 M

pH = Ka + log (0.28 / 0.32) = 4.68

b) pH = - log (0.020) = 1.70

The small amount of HCl added change the pH to more acidic (1.70) of pure water, while the pH change of the buffer is very little.

Exercises page 759

17.15

Use the information in Appendex D to calculate the pH of:

a) a solution that is 0.060 M in potassium propionate (C_2H_5COOK) and 0.085 M in propionic acid (C_2H_5COOH).

b) a solution that is 0.075 M in tri methyl amine, $(CH_3)_3N$, and 0.10 M in tri methyl ammonium Chloride $(CH_3)_3NHCl$.

c) a solution that is made of by mixing 50.0 mL of 0.15 M acetic acid and 50 mL of 0.20 M sodium acetate.

Answer:

a) $C_2H_5COOH \leftrightarrow H^+ + C_2H_5COO^-$ Weak acid (weak electrolyte)

$C_2H_5COOK \leftrightarrow K^+ + C_2H_5COO^-$ Strong electrolyte salt

	C_2H_5COOH	$\leftrightarrow$ H^+	+ $C_2H_5COO^-$
initial	0.085 M	0 M	0 M
change	- X M	+ X M	+ X M
equilibrium	(0.085 – X)M	X M	(0.06 + X)M

$Ka = 1.3 \times 10^{-5} = (X)(0.06 + X) / (0.085 - X) = 0.06\,X / 0.085$

$X = 1.842 \times 10^{-5}\,M = [H^+]$

$pH = -\log(1.842 \times 10^{-5}) = 4.73$

b) $(CH_3)_3N + H_2O \leftrightarrow (CH_3)_3NH^+ + OH^-$

$(CH_3)_3NHCl \rightarrow (CH_3)_3NH^+ + Cl^-$

	$(CH_3)_3N$ + H_2O	$\leftrightarrow$ $(CH_3)_3NH^+$	+ OH^-
initial	0.075 M	0.10 M	0 M
change	- X M	+ X M	+ X M
equilibrium	(0.075 – X) M	(0.10 + X) M	X M

$Ka = 6.4 \times 10^{-5} = (X)(0.10 + X) / (0.075 - X) = 0.10\,X / 0.075$

$X = 4.8 \times 10^{-5}\,M = [OH^-]$

$[OH^-] [H^+] = 1.0 \times 10^{-14}$ $\quad$ $[H^+] = 1.0 \times 10^{-14} / 4.8 \times 10^{-5} = 0.21 \times 10^{-9}$ M

pH = - log (0.21×10^{-9}) = 9.68

c) volume of solution = 50 mL + 50 mL = 100 mL

mole of acetic acid = (0.15 M x 50 mL) / (100 mL) = 0.075 M CH_3COOH

mole of sodium acetate = (0.2 M x 50 mL) / (1009 mL) = 0.10 M CH_3COONa

	$CH_3COOH \leftrightarrow$	CH_3COO^-	+ H^+
Initial	0.075 M	0.10 M	0 M
Change	- X M	+ X M	+ X M
Equilibrium	(0.075 – X) M	(0.10 + X) M	X M

Ka = 1.8×10^{-4} = (X) (0.10 + X) / (0.075 – X) = 0.10 X / 0.075

X = 1.35×10^{-4} M = $[H^+]$ = 1.4×10^{-4} M

<u>17.17</u>

a) Calculate the percent ionization of 0.0075 M butanoic acid (Ka = 1.5×10^{-5}).

b) Calculate the percent ionization of 0.0075 M butanoic acid in a solution containing 0.085 M sodium butanoiate.

<u>Answer:</u>

a) C_3H_7COOH butanoic acid is a weak acid

C_3H_7COONa is a strong electrolyte salt

	$C_3H_7COOH \leftrightarrow$	$C_3H_7COO^-$	+ H^+
initial	0.0075 M	0 M	0 M
change	- X M	+ X M	+ X M
equilibrium	(0.0075 – X) M	X M	X M

Ka = 1.5×10^{-5} = X^2 / (0.0075 – X) = X^2 / 0.0075

X = 3.35×10^{-4} M = $[H^+]$

Percent ionization = $H^+_{\text{equilibrium}} / [HA]_{\text{initial}}$ x 100 = (3.35×10^{-4})/ (0.0075) x 100 = 4.5%

b) $C_3H_7COOH \leftrightarrow H^+ + C_3H_7COO^-$

initial	0.0075 M	0 M	0.085 M
change	- X M	+ X M	+ X M
equilibrium	(0.0075 – X) M	X M	(0.085 + X) M

Ka = $1.5\ 10^{-5}$ = (X) (0.085 + X) / (0.0075 – X) = 0.085 X / 0.0075

X = $0.132 \times 10^{-/5}$ M = $[H^+]$

Percent ionization = (0.132×10^{-5}) x 100 / (0.0075) = 0.018%

17.21

a) Calculate the pH of a buffer that is 0.12 M in lactic acid and 0.11 M in sodium acetate.

b) Calculate the pH of a buffer formed by mixing 85 mL of 0.13 M lactic acid with 95 mL of 0.15 M sodium acetate (Ka = 1.4×10^{-4})

Answer:

a) pH = pKa + log (base / acid)

pKa = - log Ka = - log (1.4×10^{-4}) = 3.85

pH = 3.85 + log (0.11 / 0.12) = 3.85 + (-0.04) = 3.81

b) volume of solution = 85 mL + 95 mL = 180 mL

concentration of lactic acid = (0.13 M x 85 mL) / (180 mL) = 0.061 M

concentration of sodium lactate = (0.15 M x 95 mL) / (180 mL) = 0.079 M

pH = 3.85 + log (0.079 / 0.061) = 3.96

17.25

How many moles of sodium hypobromate (NaBrO) should be added to 1.0 L of 0.050 M hypobromous acid (HBrO)to form a buffer solution of pH = 9.15? assume that no volume change occur when NaBrOis added. (Ka = 2.5×10^{-9})

Answer:

pKa = - log Ka = - log (2.5×10^{-9}) = 8.60

pH = pKa log (base / acid)

9.15 – 8.60 = log (base / 0.050)

Base = [NaBrO] = 3.55 x 0.05 = 0.177 M

mole of NaBrO = 1.0 L x 0177 mol/L = 0.18 mole

17.27

A buffer solution contains 0.10 mole of CH_3COOH and 0.13 mole of CH_3COONa in 1.0 L.

a) what is the pH of this solution?

b) What is the pH of the buffer after addition of 0.02 mole KOH?

c) What is the pH of the buffer after addition of 0.02 mole of HNO_3 ?

Answer:

a) pH = pKa + log (base/acid) = 4.74 + log (0.13 mol/L)/(0.10 mol/L)

pH = 4.74 + 0.114 = 4.64

b)

	CH_3COOH	+ OH^-	$\rightarrow H_2O$ + CH_3COO^-
buffer before addition	0.10 M	0 M	0.13 M
addition	-	0.02	-
buffer after addition	(0.10 – 0.02) M	0 M	(0.13 + 0.02) M

pH = pKa + log (base / acid0 = 4.74 + log (0.15 / 0.08) = 5.01

c)

	CH_3COOH	$\leftrightarrow$ H^+	+ CH_3COO^-
buffer before addition	0.10 M	0 M	0.13 M
addition	0.020 M	0.02	0 M
buffer after addition	(0.10 + 0.02) M	0 M	(0.13 - 0.02) M

pH = pKa + log (base / acid0 = 4.74 + log (0.11 / 0.12) = 4.70

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